Carter L. Sierawski

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SUMMARY

Mechanical Engineering junior (3.96 GPA) and NASA Lunabotics Team Lead specializing in robotics and electromechanical systems. Proven experience in CAD and FEA (SolidWorks, Creo, Autodesk). Expertise in mechanical design, integrating microcontrollers, and developing algorithms for functional robotics.

EDUCATION

University of Tennessee, Knoxville Knoxville, TN
Bachelor of Science in Mechanical Engineering, GPA - 3.96 Expected May 2027
Relevant Coursework: Mechatronics, Fluid Mechanics, Thermodynamics, Mechanics of Materials, Dynamics

EXPERIENCE

Novelis - Mechanical Engineering Intern May 2025 - August 2025
- Redesigned tensioner cylinder and mount with FEA, increasing tension control by 240% and reducing stress
- Designed dual-diameter filter stand for pressure washing, improving operator workflow and ergonomics
- Retrofit squeegee roll removal cart with adjustable mounted roller reducing maintenance effort

Tickle College of Engineering Tutoring Center - Peer Learning Assistant February 2025 - present
- Tutored peers in Statics, Dynamics, and Mechanics of Materials through appointments and walk-ins
- Prepared students for exams with practice problems aligned with specific professors' course content

U.S. Engineering - Project Engineering Intern May 2024 - August 2024
- Authored Method of Procedure documentation, enabling safe execution of PCW piping chemical flush
- Created and managed punch list keeping record of outstanding tasks, ensuring adherence to project deadlines
- Coordinated maintenance of temporary cooling units and site equipment alongside QA/QC engineers

LEADERSHIP & INVOLVEMENT

Student Space Technology Association - Mechatronics Subject Matter Expert May 2025 - present
- Served as a technical mentor for electromechanical and robotic subsystems of aerospace related projects
- Evaluated and voted on project proposals as a member of the executive committee

NASA Lunabotics - Mechanical Team Lead August 2024 - present
- Designed mechanical subsystems of rover capable of excavation and berm construction in lunar regolith
- Represented the University of Tennessee at the 2025 NASA Lunabotics Competition, placing 15th of 35
- Led CAD, prototyping, and systems engineering as Mechanical Team Lead during the 2025-26 season

PROJECTS

Topology Optimized Aluminum Bracket November 2025
- Optimized a 1/8" 6061-T6 aluminum bracket for strength to weight ratio under load from a hydraulic press
- Used static and topology FEA in SolidWorks to achieve a reaction force to weight ratio of nearly 20,000
- Validated FEA with 3D printed and water-jetted prototypes, testing to failure to find stress concentrations

Robot Arm March 2025
- Designed and 3D printed a functional 4-DOF miniature robotic arm
- Integrated ESP-32 microcontroller and wireless communication to control SG90 servo motors via PWM
- Programmed inverse kinematics algorithm to convert input coordinates to smooth and repeatable movement

VEX Robotics Competition August 2017 - May 2023
- Designed and constructed various drivetrains, lifts, and subsystems to solve unique annual challenges
- Used C++ to write path planning and PID algorithms for consistent autonomous routines
- Reached World Championship in 2021 and 2023, placing 14th and 18th in division, respectively

SKILLS

Technical Skills: SolidWorks, Autodesk Fusion, Inventor, AutoCAD, Creo, FEA, GD&T, 3D Printing
Controls & Programming Languages: MATLAB, Python, C++, Arduino, ESP-32